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In [ ]: # PORTFOLIO #1: EXPRESSIONS
        # CONCEPT: SECOND LAW OF MOTION CALCULATOR
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This is a calculator used in calculating for values related in the physics concept of Newton's Second Law of Motion. Using variables, expressions, statements, and order of operations mechanisms, we can calculate for the required values.

PROBLEM 1: Calculating Mass (m) Scenario: You are helping move furniture, and you push a heavy crate with a force of 150 N. The crate accelerates across the floor at a rate of 3 m/s². What is the mass of the crate?

PROBLEM #2: Calculating Acceleration (a) Scenario: A professional bowler throws a 7 kg bowling ball down the lane. They exert a forward force of 35 N on the ball at the moment of release. How fast does the ball accelerate?

PROBLEM #3: Calculating Force (F) Scenario: An engineer is testing a new motorized cart that has a total mass of 200kg. The design requirements state that the cart must be able to accelerate from a standstill at a rate of 1.5 m/s². How much force do it need to produce?

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In [13]: # Computing for Mass
        # This sets up the variable Force and its given value for the value.
        Force = float(input("Force (N):"))

        # This sets up the variable Acceleration and its given value for the value.
        Acceleration = float(input("Acceleration (m/s^2):"))

        # This section complete the instructions given by the code and executes it.
        Mass=float(Force/Acceleration)
        # This displays the result to the user.
        print("The mass of the crate is", Mass, "kg.")
```

The mass of the crate is 50.0 kg.

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In [14]: # Computing for Acceleration
        # This sets up the variable Force and its given value for the value.
        Force = float(input("Force (N):"))

        # This sets up the variable Mass and its given value for the value.
        Mass = float(input("Mass (kg):"))

        # This section complete the instructions given by the code and executes it.
        Acceleration=float(Force/Mass)
        # This displays the result to the user.
        print("The ball accelerates at a rate of", Acceleration, "m/s^2.")
```

The ball accelerates at a rate of 5.0 m/s².

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In [16]: # Computing for Force
        # This sets up the variable Acceleration and its given value for the value.
        Acceleration = float(input("Acceleration (m/s^2):"))

        # This sets up the variable Mass and its given value for the value.
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Mass = float(input("Mass (kg):"))  
  
# This section complete the instructions given by the code and executes it.  
Acceleration=float(Mass*Acceleration)  
# This displays the result to the user.  
print("The amount of force that the motor needs to produce is", Force, "N.")
```

The amount of force that the motor needs to produce is 35.0 N.